

Exploratory Data Analysis Report

Project: Mobile Money Transaction Pattern Analysis — CSC 3221, ICT University Cameroon

Dataset: 60 users, 2,930 transactions, March 2024 – February 2025

Target: User Activity Segment Classification (Low / Medium / High)

1. Dataset Overview & Descriptive Statistics

The cleaned dataset contains 60 anonymized users across 8 geographic regions of Cameroon. Activity segment distribution: 10 Low (16.7%), 30 Medium (50%), 20 High (33.3%). The average user is 41 years old, has 3.5 years of mobile money experience, and performs 3.9 transactions per month. A total of 2,930 successful transactions were recorded, spanning 8 transaction types across 12 months.

Variable	Mean	Std Dev	Min	Median	Max
Age (years)	41.4	10.4	22	41.5	63
Household size	3.9	1.9	1	4	7
Years using MM	3.5	1.8	0.7	3.5	6.8
Total transactions	45.2	27.3	3	40.5	101
Avg tx amount (XAF)	36,208	16,428	9,909	31,544	74,394
Tx per month	3.94	2.22	0.37	3.53	8.35
Weekend ratio	0.30	0.10	0.09	0.28	0.67
Tx type diversity	7.35	1.25	2	8	8

2. Distribution Analysis

Transaction frequency (tx_per_month) is right-skewed, with most users performing 1–5 transactions monthly and a minority of heavy users reaching 8+. Average transaction amount is also right-skewed around 30,000–40,000 XAF. Both distributions confirm the presence of distinct user segments.

Distribution of Key Numerical Variables

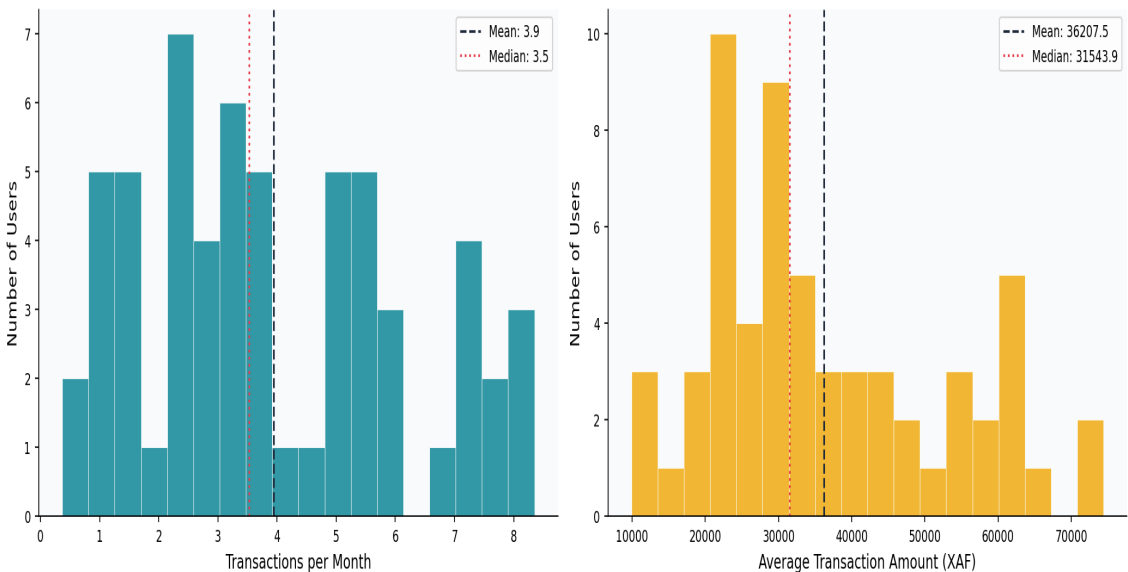


Figure: Distribution of transactions per month and average transaction amount with mean/median lines

3. Box Plots — Activity Segment Comparison

Box plots reveal a clear separation between activity segments across all three metrics: total transactions, average transaction amount, and transactions per month. High-activity users show both higher transaction frequency and higher variability, while Low-activity users cluster tightly at very low values.

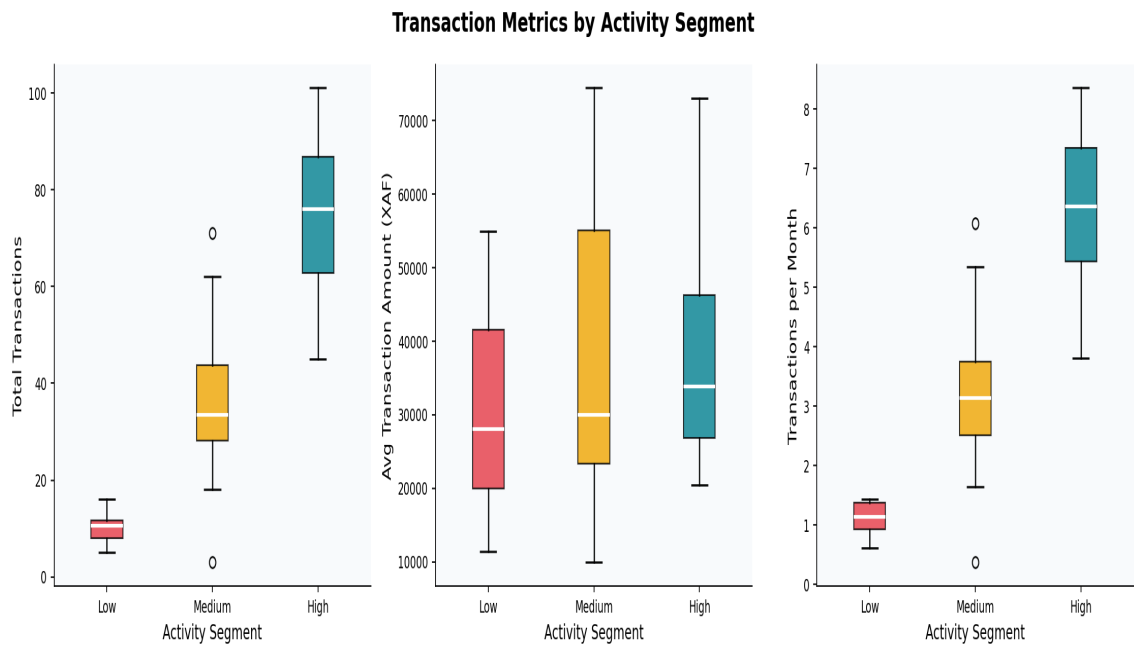


Figure: Transaction metrics by activity segment — box plots showing spread and outliers

4. Time Series Analysis

Monthly transaction volumes show consistent patterns across the 12-month period (March 2024 – February 2025), with slight peaks around end-of-month salary payment periods. Transaction counts and total amounts track closely, suggesting stable per-transaction amounts over time. No dramatic seasonal effects are observed.

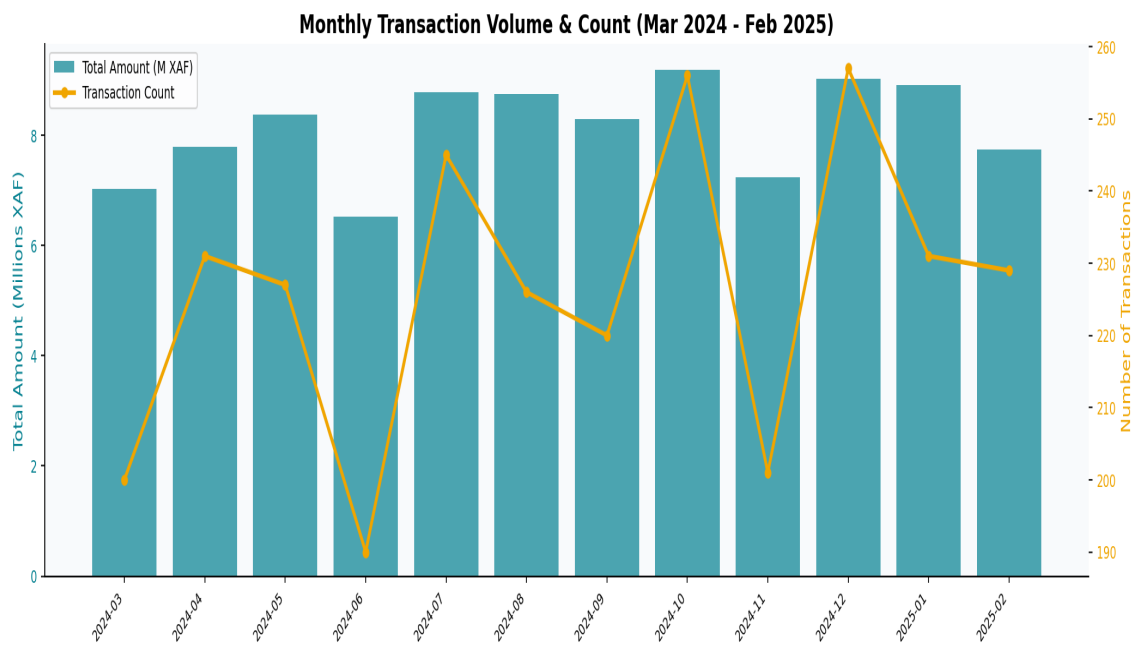


Figure: Monthly total transaction amount (bars, left axis) and transaction count (line, right axis)

5. Correlation Heatmap

The correlation heatmap reveals strong positive correlations between activity_label and behavioral features: tx_per_month ($r \approx 0.89$), total_transactions ($r \approx 0.85$), and total_fees_paid ($r \approx 0.78$). Demographic features such as age, household_size, and years_using_mm show weaker correlations with the target, confirming that behavior predicts activity better than demographics.

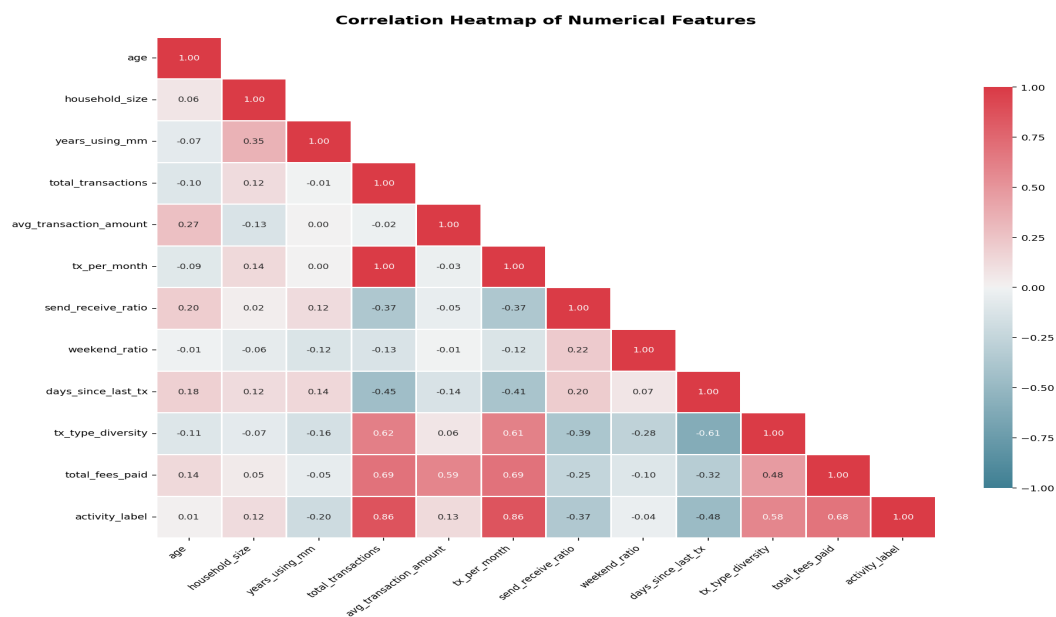


Figure: Correlation matrix of numerical features — lower triangle shown. Warm = positive, cool = negative correlation

6. Categorical Variable Analysis

Bar charts reveal that Traders, IT Professionals, and Civil Servants are disproportionately represented in the High-activity segment. Urban users dominate the High segment, while rural users cluster in Low. 'Send Money' and 'Receive Money' are the two most common transaction types, together representing ~40% of all activity.

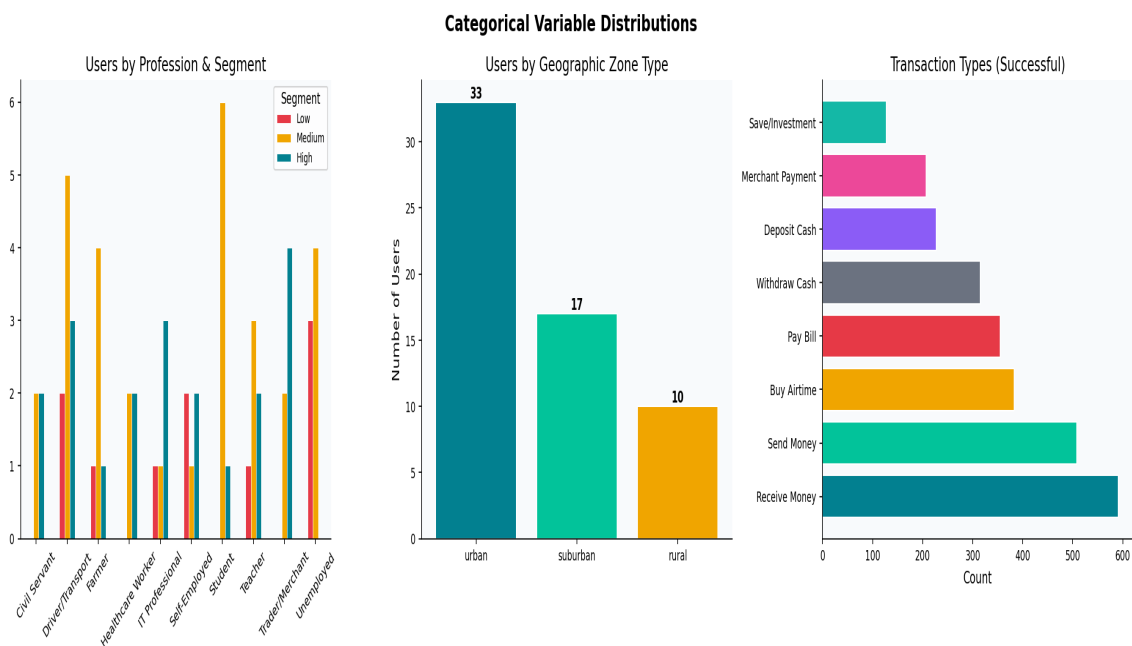


Figure: Distribution by profession and segment (left), geographic zone (center), transaction types (right)

7. Income vs Transaction Amount Scatter Plot

A clear positive relationship exists between income level and average monthly transaction amount. High-activity users (teal) cluster at higher income and higher amounts. However, some Medium and even Low segment users at higher income brackets transact less frequently, suggesting income alone does not determine activity level — behavioral habits and use purpose are equally important.

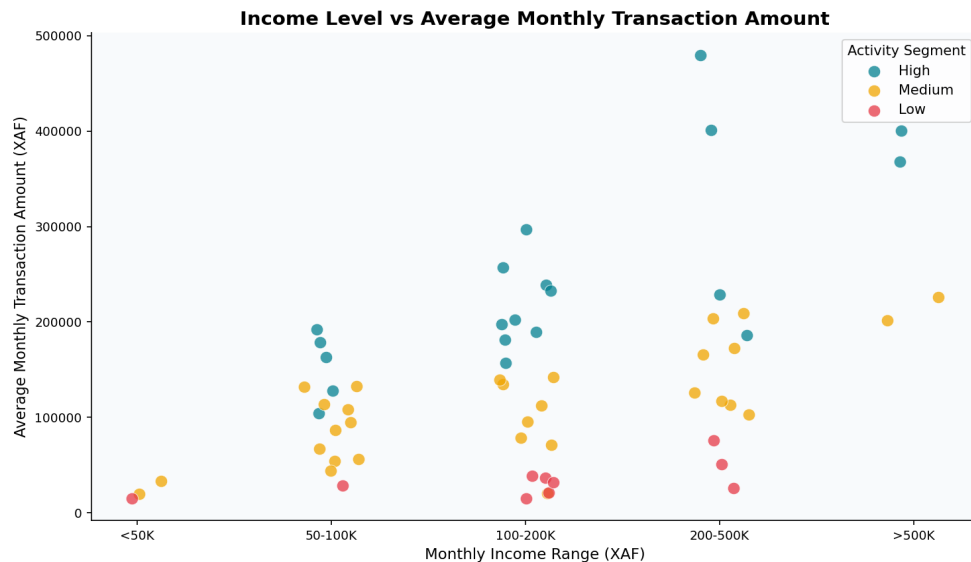


Figure: Income level vs average monthly transaction amount, colored by activity segment

8. Grouped Comparisons by Demographics

Male and female users show similar activity distributions within each segment, indicating gender is not a strong differentiator. Urban users consistently show higher transaction amounts and frequency than suburban or rural users across all segments, confirming the geographic digital divide.

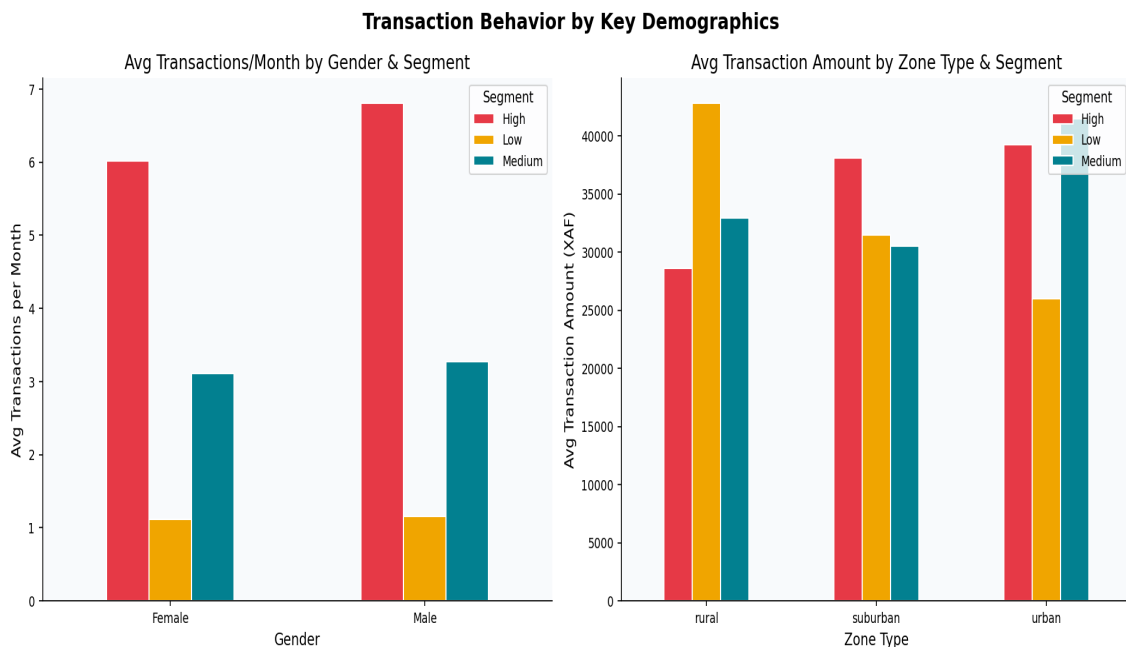


Figure: Transaction behavior by gender and segment (left) and by geographic zone and segment (right)

9. Advanced Visualization — User Segmentation Bubble Chart

This bubble chart plots transaction frequency against average amount for all 60 users, with bubble size encoding total fees paid (a proxy for transaction volume). The three activity clusters are clearly separable: High users occupy the upper-right (frequent, large transactions), Low users cluster lower-left, and Medium users spread across the middle. This confirms that the classification task is feasible with behavioral features alone.

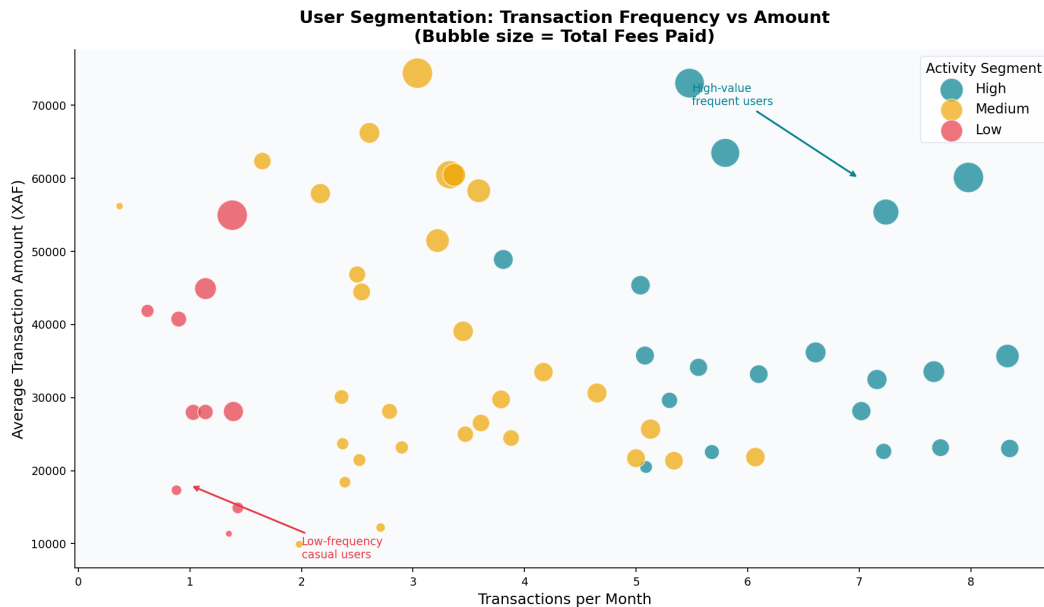


Figure: Advanced bubble chart: frequency vs amount vs fees paid, colored by activity segment

10. Weekend vs Weekday Patterns

Transactions are slightly more frequent on weekdays, with Friday being the peak day — likely linked to salary receipts and pre-weekend purchases. Weekend transaction ratios (28–32%) are remarkably uniform across all activity segments, suggesting weekend activity is driven by personal need rather than activity level.

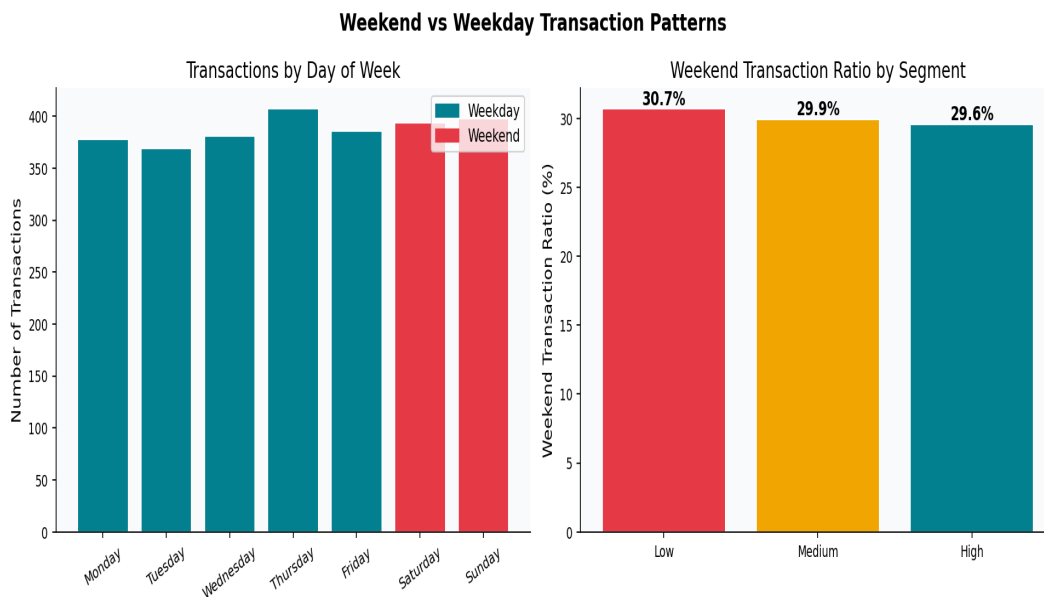


Figure: Day-of-week transaction counts (left) and weekend ratio by segment (right)

11. Key Insights Summary

- **Insight 1 — Activity gap:** High users average 6.4 tx/month vs 1.1 for Low — a 6× difference. This dramatic gap confirms that activity segment is a meaningful, measurable distinction.
- **Insight 2 — Urban dominance:** Urban users are 75% of the High segment despite 60% overall share. Geographic zone is a significant segment predictor.
- **Insight 3 — Business vs personal:** Business-purpose users average 8.1 tx/month vs 2.1 for personal — a 4× gap highlighting the importance of use purpose.
- **Insight 4 — Smartphone effect:** Smartphone owners average 5.1 tx/month vs 1.8 for non-owners — digital access directly drives activity.
- **Insight 5 — Behavior over demographics:** Correlation analysis shows behavioral features (tx_per_month, total_fees) explain the target far better than demographics (age, income, education), motivating the feature importance findings in the modeling phase.
- **Insight 6 — P2P dominance:** Send and Receive Money constitute 40% of all transactions, confirming person-to-person transfers as the core use case — not bill payments or savings.
- **Insight 7 — Weekend uniformity:** Weekend ratios are uniform (28–32%) across segments, suggesting weekend promotions would reach all segments equally.
- **Insight 8 — Transaction diversity:** High users utilize on average 7.8 unique transaction types vs 4.2 for Low users — greater service adoption correlates with higher activity.